The Link Function Problem in Psychological Interaction Testing

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Abstract

Interaction tests are common in psychology, but an interaction claim is not defined by a product term alone. It also depends on the scale on which additivity is assumed. We reframe this scale problem in the GLM language of outcome family and link function: the family describes the response, whereas the link defines the scale on which no product-term interaction is assumed. We first report a preregistered descriptive review of recent articles from five leading psychology journals, showing that interaction tests are common and that family, link, and scale-of-additivity choices are often left implicit even when outcomes have formal constraints. We develop a link-centered framework, grounded in prior work on scale dependence, outcome-model mismatch, and GLM interaction interpretation, that separates family, link, and target-metric choices. The main focus is the case in which the response family is broadly plausible but the chosen link still defines the wrong no-interaction baseline. Worked examples and simulations focus on common psychological response formats: forced-choice accuracy with a non-zero chance floor, within-family link choices such as logit versus probit, and sum scores as a related measurement case. The simulations show that plausible but inadequate links can turn additive main-effect structures on one scale into statistically significant pseudo-interactions on another. A diagnostic simulation shows that residual checks, link checks, and AIC comparisons may help, but they do not by themselves decide which scale is right for the scientific interaction claim. We argue that interaction claims are often underspecified unless researchers state, justify, and probe the scale on which no interaction is assumed.

Keywords: interactions, moderation, generalized linear models, link functions, model misspecification

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Psychological theories often test whether the effect of one variable depends on another. Stress may impair performance more strongly among people with high anxiety. A treatment may help some participants more than others. A developmental difference may be large at one age and small at another. These are interaction claims: claims about moderation, boundary conditions, and heterogeneous effects. In practice, testing such claims has become almost a routine step after main effects have been examined. This routine use is understandable, because interactions often express important theoretical ideas. Yet an interaction claim should not be interpreted automatically from the presence or absence of a product term (Rimpler et al., 2026; Rohrer & Arslan, 2021).

Statistically, many interaction claims can be understood as claims about a difference between differences. Does the treatment-control difference change across groups? Does the age-related difference differ between populations? Does the effect of task difficulty vary as anxiety increases? While this formulation sounds intuitive, it hides a crucial assumption. An interaction is a difference between differences, but whether two differences are equal depends on the scale on which those differences are expressed.

The scale problem has been recognized for a long time. Loftus argued that interactions involving response probabilities are meaningful only relative to an assumed mapping between the observed response and the theoretical component of interest (Loftus, 1978). Cox treated interaction as a statistical concept tied to model formulation, transformation, and interpretation (Cox, 1984). Wagenmakers and colleagues later emphasized the distinction between removable and nonremovable interactions, showing that some interactions depend on the measurement scale and therefore do not support unambiguous conclusions about latent psychological processes (Wagenmakers et al., 2012). Busemeyer and Jones (1983) showed how conclusions about multiplicative combination rules depend on the measurement model and on measurement error in the observed variables. Lubinski and Humphreys illustrated how apparently substantive moderator effects can arise from model specification choices rather than real psychological synergy (Lubinski & Humphreys, 1990). More recently, Rohrer and Arslan argued that interaction tests can provide precise answers to vague questions when researchers do not specify the estimand, scale, or substantive meaning of the interaction (Rohrer & Arslan, 2021).

In psychology, this scale problem often has a measurement origin. Researchers usually reason about constructs such as ability, symptom severity, attitude strength, processing efficiency, or others as if they were continuous underlying dimensions. Most psychological dimensions may indeed be treated as smooth and roughly Gaussian, because they plausibly reflect many small sources of variation. If such processes were observed directly on an interval scale, a Gaussian identity model might often be a reasonable solution. But psychological phenomena are rarely observed directly. They are observed indirectly through tasks, items, ratings, counts, accuracies, and summed responses. These observations impose bounds, thresholds, discreteness, floors, ceilings, and chance levels. The link-function problem is one formal expression of this broader psychometric problem: the model must say how an additive structure on one scale is mapped onto the scale on which the data are observed.

This is the reason for the emphasis on binary, binomial, ordinal, and bounded examples in this paper. Many standard psychological outcomes begin as discrete or ordered response processes, even when they are later reported as percentages, means, totals, or composites: a trial is correct or incorrect, an item is endorsed or rejected, a respondent chooses one ordered category, and a score aggregates such responses. Response times, psychophysiological measures, neuroimaging indices, movement measures, and other continuous behavioral traces raise related issues with positive support, skew, or transformations. Still, bounded and ordinal response formats are one of the main routes through which psychological processes become observed data.

A related literature has sharpened the problem for modern psychological data. Work on multiplicative interaction models has warned that interaction estimates can depend strongly on functional-form assumptions and on the observed range of the data (Hainmueller et al., 2019). Recent guidance on power and stability also shows that interaction estimates are often fragile for ordinary statistical reasons, such as small effects, low reliability, and limited power (Baranger et al., 2023; Castillo et al., 2026). McCabe and colleagues showed that, in generalized linear models

(GLMs) for probabilities and counts, product-term coefficients are not generally equal to interaction effects on the response scale (McCabe et al., 2022). Work on nonlinear probability models reaches the same practical conclusion: interaction effects should often be evaluated as conditional marginal effects or response-scale contrasts, not as product coefficients alone (Ai & Norton, 2003; Mize, 2019). More broadly, estimand-centered approaches recommend treating models as tools for generating substantively meaningful quantities rather than interpreting coefficients mechanically (Rohrer & Arel-Bundock, 2026). Together, these contributions show that interaction testing requires more than adding the right-hand-side product term to a regression equation.

Some methodological advances consist in taking old warnings seriously enough to make them operational. This paper applies that idea to interaction testing in GLMs. The scale dependence of interactions is already well established (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012), and product terms in GLMs cannot be read directly as response-scale interactions (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019). Domingue and colleagues showed that outcome-model misspecification can produce false interaction findings for binary, count, and censored outcomes (Domingue et al., 2024). Liddell and Kruschke showed that treating ordinal responses as metric can affect error rates and interaction conclusions (Liddell & Kruschke, 2018). Czado and Santner, and later Yu and Huang, studied the inferential consequences of link misspecification in binary GLMs and GLMMs (Czado & Santner, 1992; Yu & Huang, 2019). Rimpler and colleagues showed that product terms in linear models can mismatch the functional form implied by a substantive conditional-effect claim (Rimpler et al., 2026).

These contributions identify adjacent parts of a shared scale problem. The present paper narrows the problem to interaction-focused link choice in applied psychological analyses, especially cases where the outcome family is broadly plausible but the link still defines a disputable no-interaction baseline. The incremental contribution is to separate outcome family, link function, and target metric in one framework for interaction testing, then turn this separation into an applied workflow for common psychological response formats. The review documents how often this residual ambiguity persists in recent psychological practice. The worked examples and simulations quantify the inferential risk in selected cases where link choices can generate pseudo-interactions.

Choosing the outcome family solves only one part of interaction testing. Two models can respect the same outcome family while defining different no-interaction baselines. For response times, for example, a Gamma family may be plausible because the outcome is positive, continuous, and right-skewed. Yet `Gamma(link = "log")` assumes additivity in log mean response times, which corresponds to proportional change on the millisecond scale, whereas `Gamma(link = "inverse")`, the canonical link, assumes additivity in inverse mean response times. The relevant question is whether the scientific claim concerns proportional change, reciprocal change, absolute change, or some other scale. Link misspecification can affect likelihood-based inference in GLMs and GLMMs (Czado & Santner, 1992; Yu & Huang, 2019).

The same logic applies to binary and accuracy data, which are frequently reported in psychological research. A binomial family may correctly represent the sampling process of correct and incorrect responses, but the link still defines the scale of additivity. A standard logit link maps the linear predictor to the full 0-to-1 probability range, and a probit link makes a similar but not identical mapping, with different curvature especially away from the middle of the probability scale. In a two-alternative forced-choice task, however, the task-implied lower bound for performance above guessing is often 0.50. A chance-corrected link can encode this lower asymptote directly when that is the scale required by the scientific claim. Thus, the distinction is not only between linear models and GLMs. It is also across plausible links within a generally appropriate family.

Figure 1 shows why link choice is a scale choice, not only a curve-fitting choice. In each panel, equal steps on the target-link scale map to unequal steps on any other scale. Floor and ceiling effects are the most visible version of this problem, but the same logic applies whenever the mapping from the linear predictor to the response is nonlinear. This is the link-scale version of the scale-dependence problem introduced above.

Figure 2 uses a simulated two-alternative forced-choice accuracy dataset as an illustrative example. Each participant completes 50 trials, and the outcome is the proportion correct. The predictors are age and group. The data-generating model includes an age effect and a group effect, but no age-by-group term on the chance-corrected logit scale, the scale on which the task has a 0.50 chance floor. The figure applies the same link-scale point to a concrete moderation claim: the same data can appear to support different moderation conclusions when fitted models define different no-interaction baselines (Cox, 1984; Domingue et al., 2024; Loftus, 1978; McCabe et al., 2022).

Some interactions remain stable across scale choices, especially non-removable or crossover interactions (Cox, 1984; e.g., Loftus, 1978; Wagenmakers et al., 2012). Many quantitative interaction claims remain underspecified when researchers leave implicit the scale of additivity, the distinction between linear-predictor and observed-response scales, or the robustness of the conclusion across plausible links. In this sense, link choice becomes part of the scientific claim.

The paper proceeds in four steps. First, the preregistered review describes current reporting practice: how often interaction tests are used, how often family and link choices are explicit, and how often outcomes have formal constraints that make identity-link additivity a modeling assumption requiring justification. Second, the framework separates outcome family, link function, and target metric, with special attention to cases in which the family is plausible but the link still sets the wrong additivity scale. Third, the applied sections develop three psychology-relevant cases: forced-choice accuracy with a non-zero chance floor, sum scores, and within-family link choice. Fourth, the simulations quantify when link or metric choices can generate pseudo-interactions, and the diagnostic example asks what standard checks can and cannot detect. The practical starting points summarized later in Table 8 are one intended output of this framework, and the sections that follow provide the rationale behind those recommendations.

Current Practice in Psychological Interaction Testing: A Preregistered Review

Why This Review Matters

The review examines whether the scale-dependence problem identified above is visible in current reporting practice. We operationalized this through four questions: how often do empirical articles test interactions, how often do interaction-testing articles use non-identity links, how often is the link function explicitly reported, and which response-variable types appear in interaction analyses. The review is descriptive. It does not classify individual interactions as substantively valid, removable, or nonremovable, because this judgment often requires information that published reports rarely provide, such as complete model coefficients, interaction plots, the assumed measurement scale, and the substantive scale on which the interaction was meant to hold. This limitation is important because the removable/nonremovable distinction is central in the classical scale-dependence literature (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012). The purpose of the review is narrower: to document whether the family, link function, and scale of additivity are made explicit enough for readers to understand which interaction claim was tested (Mustillo et al., 2018).

Review Method

The review protocol was preregistered before the records were screened, at: https://osf.io/bdu73/ We specified a descriptive review of recently published psychological research articles, with four primary goals: estimating how often empirical articles test interactions, how often interaction-testing articles use non-identity links, how often the link function is explicitly reported, and which response-variable types are used in interaction analyses. The protocol defined a descriptive review of current practice across major areas of psychology, not a census of all journals or publication years.

Sampled articles were recent publications, all from 2024, in five pre-specified reference journals across major areas of psychology: *Psychological Science*, *Journal of Experimental Psychology: General*, *Journal of Personality and Social Psychology*, *Developmental Science*, and *Psychological Medicine*. These journals were chosen to cover general, experimental, social, developmental, and clinical psychology. The sampling design was targeted and descriptive. It was not intended as a random or stratified sample of all psychology journals. Its purpose was to characterize reporting practice in outlets whose standards and reach make them broadly influential beyond the sampled year. Eligible articles were empirical articles presenting original psychological research. Reviews, meta-analyses, editorials, theoretical articles without empirical data, qualitative-only articles, replication studies, and psychometric validation studies were excluded. The preregistered coding scheme targeted observed-outcome analyses and excluded articles in which the relevant tests were conducted only at the latent-variable level, or in which the empirical analyses relied only on structural equation models (SEM). This scope decision kept the review focused on observed outcomes analyzed with regression, ANOVA, or GLM-like models. The screening-flow table records 2 SEM or latent-variable-only exclusions, so this decision affected only a small part of the screened set. SEM is relevant to the broader issue because it can make the measurement model explicit. It also places the link-function problem on a different level, involving thresholds, loadings, and latent-response metrics. Evaluating such models with the observed-outcome coding scheme would have required additional preregistered coding rules. SEM therefore falls outside the scope of this review and would benefit from a dedicated preregistered protocol. Articles were then coded in detail if they tested at least one interaction effect, either through statistical modeling or analysis of variance.

For each interaction-testing article, coders recorded whether at least one tested interaction used a non-identity link function, whether the link function was explicitly reported, whether at least one interaction involved a formally incorrect identity-link analysis of a constrained observed outcome, whether at least one such interaction was statistically significant or interpreted analogously, and which response-variable types were analyzed. We use the term formally incorrect identity-link analysis of a constrained observed outcome for cases in which an interaction was analyzed with an identity link although the reported outcome had formal constraints, such as bounds, discreteness, positivity, or a known chance level. In these cases, the identity link is formally incorrect as a generative model for the observed outcome, unless it is explicitly justified as an approximation or as the target observed-score metric. This category is descriptive and conservative. It does not imply that substantive conclusions are necessarily false. It is a screening flag. It does not grade severity. It groups heterogeneous cases, from mild observed-scale approximations far from the bounds to analyses where the outcome constraints make identity-link additivity much harder to defend. The coding therefore documents the potential scale of the reporting problem, while the worked examples and simulations show mechanisms by which such choices can matter. The protocol specified descriptive summaries, including proportions and Wilson 95% confidence intervals for Boolean variables. Double-coded records with complete coder pairs showed high agreement, and the same information is reported together with Cohen's kappa: 92.0% agreement, kappa = 0.81 for non-identity link use, 88.9% agreement, kappa = 0.66 for explicit link reporting, 88.3% agreement, kappa = 0.71 for formally incorrect identity-link analyses of constrained observed outcomes, and 91.5% agreement, kappa = 0.68 for significant interactions among these cases. We report raw agreement and kappa together because several coded variables had imbalanced marginal distributions. The coder-1/coder-2 yes rates were 28.1% / 30.2% for non-identity link use, 20.6% / 20.6% for explicit link reporting, 71.6% / 72.1% for formally incorrect identity-link analyses, and 86.5% / 82.3% for significant interactions among these cases. These marginal rates document why kappa can be conservative in this setting, while still allowing readers to evaluate the agreement pattern directly. The full review protocol, coding definitions, and extended descriptive tables are reported in the Supplement.

Review Findings

Three descriptive results organize the review findings. First, interaction testing was common. The review identified 331 eligible empirical articles, and 200 of these articles tested at least one interaction, corresponding to 60.4%, 95% CI [55.1, 65.5]. This confirms that interaction testing is a routine part of empirical reporting in the sampled journals.

Second, model-scale reporting was uncommon. Among the 200 interaction-testing articles, 58 used at least one non-identity link function, 29%, 95% CI [23.2, 35.6], and only 46 explicitly reported the link function, 23%, 95% CI [17.7, 29.3]. Thus, in about 77.0% of interaction-testing articles, the link function was not explicitly stated. This is the main reporting problem documented by the review: the scale on which no interaction is assumed is often left implicit.

Third, constrained outcomes were often analyzed with identity-link models. Among the 200 articles testing interactions, 144 included at least one identity-link analysis in which the reported outcome had formal constraints that make identity-link additivity formally incorrect as a generative model unless explicitly justified as an approximation or as the target observed-score metric, 72%, 95% CI [65.4, 77.8]. This proportion is a conservative screening count. It is not an estimate of false substantive conclusions, and it is not a severity rating. The category uses transparent screening and avoids retrospective case-by-case severity judgments that published reports often could not support. Cases were flagged only when the formal constraint was identifiable from the published report. When the available report did not allow us to establish this mismatch, the case was not counted. The review result shows that many published reports did not make the family, link, and scale-of-additivity choices explicit enough for readers to evaluate the interaction claim. This is the same broad concern raised by work on outcome-model mismatch and metric assumptions in interaction testing (Domingue et al., 2024; Liddell & Kruschke, 2018). Among these 144 cases, 124 reported at least one statistically significant interaction, 86.1%, 95% CI [79.5, 90.8]. In absolute terms, this corresponds to 124 of the 200 interaction-testing articles, or 62.0%. This last number should not be read as a false-positive estimate. It records how often formally incorrect identity-link analyses appeared in reports where statistically significant interaction claims could matter for interpretation. Table 1 summarizes these descriptive results.

The pattern was present across the sampled journals. In the supplementary by-journal table, the proportion of formally incorrect identity-link analyses of constrained observed outcomes among interaction-testing articles ranged from 40.0% in Psychological Medicine to 95.0% in Psychological Science. In this sample, Psychological Medicine had the lowest screening proportion and the highest non-identity-link proportion (40.0%). This journal, which uses many outcomes for which non-identity links are standard, contributed the fewest formally incorrect identity-link cases. This pattern also shows that the overall estimate was not driven by the clinically oriented journal, where non-identity links were more common.

The outcomes used in interaction analyses were heterogeneous. The most common broad, non-exclusive categories were binary, accuracy, or proportion outcomes (84 articles), sum scores or composites (66 articles), ordinal, Likert, or rating outcomes (36 articles), response times or durations (30 articles), neural or physiological measures (23 articles), difference or distance scores (17 articles), counts or error counts (9 articles), and correlations or associations (8 articles). This heterogeneity matters because the identity link has different implications across outcome types. It also shows why the worked examples focus on bounded, binomial, ordinal, and aggregated responses. Many observed psychological variables are built from binary or ordered response events, then reported as proportions, ratings, averages, totals, or composites. For some approximately continuous outcomes, additivity on the observed scale may be a defensible target. For bounded, discrete, ordinal, chance-constrained, or heavily transformed outcomes, however, additivity on the observed scale is an assumption that should usually be stated and examined. This is especially clear for ordinal and Likert-type outcomes, where treating ordered categories as metric can affect estimates, error rates, and interaction conclusions (Bürkner & Vuorre, 2019; Liddell & Kruschke, 2018).

What the Review Shows, and What It Does Not Show

The review documents a reporting gap, not an error rate. Identity-link analyses may be defensible as approximations or direct observed-scale tests, especially when the target estimand is the observed score. This justification should be explicit when the outcome has known constraints. The review also does not adjudicate removability at the article level. It shows a narrower point: in leading psychology journals, interaction testing is common, while the scale on which no interaction is assumed is often left implicit.

The next section formalizes why this matters. We distinguish the outcome family from the link function, show why choosing the family is not enough for interaction testing, and separate interactions on the linear-predictor scale from interactions on the observed response scale.

The Link Function Problem

The review described a reporting problem: interaction tests are often presented without an explicit statement of the outcome family, link function, or scale of additivity. We now formalize why this matters. In GLMs, the link function defines the model scale on which effects are assumed to add. This matters especially when psychological data are indirect observations of a target process, because a task or scoring rule can remap that process onto a constrained scale.

In a generalized linear model (GLM), the expected outcome for observation i, denoted $\mu_i = E(Y_i)$, is related to a linear predictor through a link function (McCullagh & Nelder, 1989; Nelder & Wedderburn, 1972):

$$g(\mu_i) = \eta_i.$$

With two predictors, X_i and Z_i, and their product term, the linear predictor is written as

$$\eta_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i.$$

The first equation contains the link function, $g(\cdot)$. This is the function that maps the conditional mean of the outcome, μ_i, onto the linear predictor, η_i. The second equation then specifies what is additive on that linear-predictor scale: the intercept, the two main effects, and, if included, the product term.

This separation is important because a GLM involves two related but different choices. First, the analyst chooses a family, that is, a model for the kind of outcome being analyzed. This choice defines the possible values of the response and its sampling process, for example Gaussian scores, binomial successes, Poisson counts, or positive continuous outcomes. Second, the analyst chooses a link function, that is, the function $g(\cdot)$ that defines the scale on which the predictors are combined additively.

This second choice is crucial for interaction testing. In a model without the product term, X_i and Z_i are additive on the η_i scale, not necessarily on the observed outcome scale. The link function defines how this additive structure is translated back to the response scale. For this reason, choosing a plausible family is necessary but not sufficient. The family can keep predictions within the right range, but the link still defines the scale on which the product term is interpreted as a departure from additivity.

What the link function does

The linear predictor, η, is the model's additive scale. Intercepts, main effects, product terms, covariates, and random effects are combined before the inverse link is applied. Many psychological outcomes, however, are not direct readings of the psychological continuum of interest. They are observed through constrained measurement scales. Counts cannot be negative. Probabilities must lie between 0 and 1. Forced-choice accuracy may have a theoretical floor above 0, for example .50 in a true/false task. Response times are positive and often right-skewed. Sum scores have lower and upper bounds, and are often discrete.

The inverse link, $g^{-1}(\eta)$, maps the additive model back to the expected response. This mapping is not neutral for interactions: the link function defines what "no interaction" means in the fitted model.

The distinction is hidden in ordinary linear models because the identity link makes the two scales coincide. If $g(\mu) = \mu$, then additivity on the linear predictor scale is also additivity on the response scale. For any other link function, this equivalence disappears. A logit link defines additivity in log odds. A probit link defines additivity on a latent-normal scale. A log link defines additivity in log means (which corresponds to multiplicative change on the response scale). The same general point applies within a family. As noted in the introduction, a Gamma model for response times may use an identity, log, or inverse (the canonical) link: the family is the same, but the implied no-interaction baseline is different across link choices.

A simple count example

Suppose children make fewer errors as they grow up. A Poisson model with a log link can represent this as a linear age effect on the log mean,

$$\log\{E(Y)\} = \beta_0 + \beta_1\text{age},$$

with $\beta_1 < 0$. On the observed count scale, the same model implies $E(Y) = \exp(\beta_0 + \beta_1\text{age})$, a curved decreasing function. The curvature is not an extra psychological mechanism, but the expected consequence of mapping an additive model on the log scale back to non-negative counts.

Now add a group/condition difference in overall error rate, with no age-by-condition product term on the log scale. On the observed count scale, the absolute age-related reduction will usually be larger where expected errors are high and smaller where expected errors are low. A model that assumes additivity on another scale, especially an identity link, can treat this changing absolute difference as moderation. The same logic applies within the same family: a Poisson model with an identity link and a Poisson model with a log link both model counts, but they define different no-interaction baselines. A Poisson-log model is appropriate when log-count additivity is the target. A product term is a departure from additivity on the fitted link scale, whereas response-scale differences may change even when there is no product term on the generating scale (Domingue et al., 2024; McCabe et al., 2022).

The same logic applies directly to interaction coefficients. The coefficient β_3 is a product-term interaction on the link scale, $g(\mu)$. When $\beta_3 = 0$, the fitted model assumes additivity on that scale. It does not necessarily assume additivity on the observed response scale. A response-scale interaction can instead be defined as a difference between expected differences. For two binary predictors, one such quantity is

$$\{E(Y \mid X = 1, Z = 1) - E(Y \mid X = 0, Z = 1)\} - \{E(Y \mid X = 1, Z = 0) - E(Y \mid X = 0, Z = 0)\}.$$

In a nonlinear GLM, this quantity depends on the inverse link, $g^{-1}(\eta)$. It can be nonzero when the product-term coefficient is zero, and it can differ in size or sign from the product-term coefficient when the product term is nonzero. This is why product-term coefficients in GLMs should not be treated as automatic answers to response-scale moderation questions (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019).

One way to see the mechanism is to write it in terms of curvature. Suppose that the model is additive on the link scale,

$$\eta = \alpha + \beta X + \gamma Z, \qquad \mu = h(\eta),$$

where $h = g^{-1}$ is the inverse link. There is no product term on the link scale. With continuous predictors, however, the response-scale cross-partial derivative is

$$\frac{\partial^2 \mu}{\partial X \partial Z} = h''(\eta)\beta\gamma.$$

Thus, an additive model on the link scale can imply a response-scale interaction whenever the inverse link is locally curved. In a 2×2 design, the same point appears as a difference between differences:

$$h(\eta + a + b) - h(\eta + a) - h(\eta + b) + h(\eta) \approx h''(\eta)ab.$$

The size of the induced response-scale interaction therefore depends on the local curvature of the inverse link and on the size of the two main effects. It is most visible when expected values fall in regions where the inverse link compresses response-scale change differently across groups or conditions, for example when one condition is close to a floor, ceiling, chance level, or tail, while another lies closer to the middle of the scale.

The link-function problem is therefore a scale-of-additivity problem. An interaction claim is always relative to a scale: it says that the effect of X is not constant across levels of Z when effects are expressed on that scale. An identity-link model evaluates this claim on the expected observed mean. A binomial-logit model evaluates it in log odds. A binomial-probit model evaluates it on a probit scale. A chance-corrected binomial link evaluates it on a scale that maps the linear predictor onto the interval from chance performance to perfect performance. These are related statistical models, but they do not express the same scientific null.

This point helps separate three analytic choices that are often mixed together. The first choice is the outcome family: does the model respect the support and sampling process of the observed outcome? The second choice is the link, which defines the scale on which effects are additive when the product term is zero. The third choice is the target metric: is the observed score itself the scale on which the theoretical interaction claim is meant to hold? These choices can overlap in practice, but they are conceptually distinct. The focus of this paper is the second choice, especially cases in which the family is broadly plausible but the scale of additivity remains underjustified.

The three rows in Table 3 should not be read as mutually exclusive error types. They are analytic decisions that can be separated for clarity, but they often co-occur in real analyses. Sum scores are the clearest boundary case: the manifest score is already a measurement mapping from item responses, and any model fitted to that score also chooses a link or additivity scale. The distinction is useful because it asks which part of the analysis carries the scale assumption: the sampling model, the link used for a given response, or the decision to treat the manifest score as the target metric.

Our paper focuses mainly on the link/additivity-scale choice, while keeping family and target-metric choices visible. This focus is narrower than the general warning that linear models can be inappropriate for non-Gaussian outcomes. Researchers may choose a broadly appropriate family and still leave the scientific scale of additivity unspecified. For instance, a binomial family may correctly represent successes out of trials, but it does not choose among logit, probit, complementary log-log, flexible generalized links, or a chance-corrected link with a lower asymptote (Aranda-Ordaz, 1981; Stukel, 1988). Link misspecification can affect inference in binary GLMs and GLMMs, and diagnostic evidence may not identify a unique correct link (Czado & Santner, 1992; Yu & Huang, 2019). A separate target-metric choice concerns whether the claim is about the manifest score, such as raw sum-score units, or about an underlying construct scale that requires an explicit measurement model.

The worked examples use the three choices in different ways. The forced-choice case holds the binomial family fixed and asks whether the link should encode the task-implied chance floor. The logit-probit case holds both the binary family and the 0-to-1 bounds fixed, then asks whether two plausible links define the same no-interaction baseline for the interaction claim. The sum-score case places the target metric in the foreground, because the manifest total score is already a measurement mapping from item responses to a bounded scale. The three cases are complementary. Together, they cover the main routes through which scale assumptions enter interaction tests in psychological research.

A link-aware interaction analysis therefore begins with four questions. First, what is the estimand: a link-scale coefficient, a probability difference, a marginal effect, a risk ratio, a count difference, or another response-scale contrast? Second, what constraints does the outcome impose, such as bounds, discreteness, ordinal categories, skew, ceilings, floors, or chance performance? Third, which scale of additivity is scientifically plausible for the process under study? Fourth, does the substantive conclusion survive plausible alternative links? These questions do not make interaction testing automatic, but they make the assumptions visible enough to evaluate.

This framework also clarifies the role of the simulations that follow. The simulations ask whether an interaction can be generated, removed, or distorted when the fitted model assumes additivity on a different scale from the data-generating process. This matters because the examples below are common routes from psychological processes to observed data. Accuracy, forced-choice responses, Likert ratings, item endorsements, symptom counts, and many sum scores are generated by bounded, binary, ordinal, or aggregated response formats. Response times and physiological or neuroimaging measures raise other link questions, especially because they are positive, skewed, or constrained. The next sections therefore focus on forced-choice accuracy with chance performance, sum scores, and within-family link choices for binary or binomial responses.

Box 1. Prediction, Explanation, and the Scale of Interaction

The importance of link choice depends on the goal of the analysis. For purely predictive purposes, the primary criteria may be predictive accuracy, calibration, generalizability, and decision usefulness. A model can be useful if it predicts future observations well, even if its coefficients do not provide a faithful description of the psychological process that generated the data.

Scientific interaction claims require a different standard. When researchers claim that an effect is moderated, they usually claim more than that one model predicts slightly better than another. They claim that the effect of one variable changes across levels of another variable in a way that is meaningful for theory. That claim requires an estimand. It also requires a scale. A treatment effect might be larger in absolute probability points, larger in odds ratios, larger on a latent response scale, or larger only after a transformation. These are different claims, and they can lead to different conclusions.

This distinction is consistent with estimand-centered workflows that treat models as tools for generating clear target quantities rather than as collections of coefficients to be interpreted mechanically (Rohrer & Arel-Bundock, 2026). For interaction testing, the relevant target quantity should be stated before the product term is interpreted. If the target is a response-scale difference-in-differences, the analyst should compute and report that quantity. If the target is additivity on a latent or link scale, the link function should be justified as part of the scientific model.

Prediction and explanation remain connected in practice: predictive checks, calibration, and out-of-sample performance can all inform scientific modeling. Good predictive fit does not specify the scale on which a moderation claim is made. For scientific inference about interactions, the link function is part of the theoretical specification, not only a computational convenience (Rohrer & Arel-Bundock, 2026).

Forced-Choice Accuracy and the Non-Zero Chance Floor

Forced-choice accuracy is a useful case because it separates the outcome family from the link scale. If a participant completes k_i trials and answers Y_i trials correctly, the binomial sampling model is often the natural starting point:

$$Y_i \sim \text{Binomial}(k_i, p_i),$$

where p_i is the expected probability of a correct response. This choice concerns the sampling process. It does not, by itself, decide the lower bound that is meaningful for the psychological task. In a 2-alternative forced-choice task, or in a true/false task when random responding between two alternatives is the relevant chance process, guessing implies an expected accuracy of .50. More generally, in an m-alternative forced-choice task, the chance level is $c = 1/m$. For these tasks, the expected performance curve may have a lower asymptote at chance, rather than at 0, when the target claim concerns performance above guessing.

This distinction gives three increasingly specific specifications. The weakest is a Gaussian identity model fitted to proportions. It ignores the binomial trial process and treats accuracy as an approximately continuous outcome. It also assumes additivity directly in percentage-correct units and can generate fitted values outside the admissible 0-to-1 range. This model may sometimes be useful as a rough descriptive approximation, especially far from the boundaries, but it does not encode either the trial structure or the task-implied lower bound.

A standard binomial logit or probit model improves on this by respecting the binomial sampling process and constraining fitted probabilities to the interval from 0 to 1:

$$p_i = F(\eta_i),$$

where F is the logistic cumulative distribution function for a logit link, or the standard normal cumulative distribution function for a probit link. This model is right about the binomial structure of the data. However, in a forced-choice task with a known chance level, it still defines the lower asymptote as 0. A standard binomial model can respect the sampling process while encoding a lower asymptote that is not aligned with the task-based interpretation of performance above chance.

A chance-corrected binomial link keeps the binomial sampling model but maps the linear predictor to the interval from chance to 1:

$$p_i = c + (1 - c)F(\eta_i).$$

For a 2-AFC or true/false task with chance level .50, this becomes

$$p_i = .50 + .50F(\eta_i).$$

Equivalently, the link is applied to the above-chance quantity $(p_i - c)/(1 - c)$ rather than to the raw probability p_i. The no-interaction baseline is therefore different. A product term in a standard binomial-logit or binomial-probit model tests additivity on a scale defined over the full 0-to-1 probability range. A product term in a chance-corrected binomial model tests additivity on a latent performance-above-chance scale that already encodes the task-implied lower asymptote. These are different modeling claims, even though both use a binomial family.

The forced-choice case is a direct example of this curvature problem. The difference is easiest to see when one group, condition, or region of the predictor space lies close to chance. Suppose that age has the same effect in two groups on the latent performance-above-chance scale, with no true age-by-group product term on that scale. On the observed accuracy scale, the lower-performing group can still show stronger compression near .50, because the chance-corrected curve has less response-scale room close to its lower asymptote. A standard logit or probit link places this compression near 0 rather than near .50. The fitted model can then use an age-by-group product term to absorb the curvature created by the misplaced lower bound. The issue extends beyond obvious floor effects: it arises whenever candidate links allocate different response-scale room to the same link-scale change across the predictor range. The resulting interaction is therefore not merely random Type I error. It is a pseudo-interaction produced by a mismatch between the scale used by the model and the scale implied by the task. This is consistent with broader work showing that outcome-model misspecification can create false discoveries in interaction analyses (Domingue et al., 2024), with work showing that product-term coefficients in nonlinear probability models are not automatically response-scale interaction effects (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019), and with evidence that binary-link misspecification can affect likelihood-based inference (Czado & Santner, 1992; Yu & Huang, 2019).

Standard binomial models remain useful for many accuracy analyses. The chance-corrected link is most defensible when the chance level is known from the task design, guessing is a plausible lower-bound process, and below-chance performance is not substantively meaningful for the target claim. In the simplest case, the chance level can be fixed by design, for example $c = 0.50$ in a 2-AFC task. In other cases, analysts may estimate the lower asymptote or report sensitivity analyses over plausible values of c, although estimating it requires enough information and stronger assumptions. When a chance-corrected model is unstable or fails to converge, the practical response is to simplify the specification and report the identification problem transparently. Analysts can fix c from the task design, remove unsupported random slopes, or fit a simpler trial-level or aggregate model. They can then compare standard and chance-corrected predicted curves and report response-scale contrasts for the target interaction. If the design does not identify the lower asymptote well enough, the result should be presented as link-conditional, without forcing the chance-corrected model to deliver a single answer. Richer models may also be needed if there are lapses, response biases, item heterogeneity, learning across trials, systematic below-chance performance, or participant-specific guessing strategies. The practical question is whether the link used for the interaction test encodes the scale on which no interaction is scientifically meaningful.

The chance-corrected GLM link is closely related to psychometric-function models, where lower asymptotes, guessing, lapses, and goodness-of-fit checks are part of the response model, and to item response models that include a lower asymptote (Knoblauch & Maloney, 2012; Wichmann & Hill, 2001). For example, the 3-parameter logistic IRT model includes a guessing parameter at the item level, rather than treating the chance floor as a single aggregate constraint. When item-level or trial-level data are available, and when the target claim concerns latent performance rather than an aggregated accuracy proportion, such measurement models or trial-level GLMMs may be more appropriate than a single aggregate GLM (Moscatelli et al., 2012). The chance-corrected link used here should therefore be understood as a GLM-level representation of a design constraint. It is not a replacement for item-level measurement models when the data and the scientific question call for them.

The same dataset can also be analyzed for different purposes. If the goal is prediction, standard and chance-corrected binomial models can be compared by predictive accuracy, calibration, and generalizability. If the substantive estimand is an observed-scale quantity, such as a difference in percentage correct or a difference between such differences, that quantity should be reported directly as a response-scale contrast. If the claim concerns latent performance above chance, ability, sensitivity, or propensity to answer correctly beyond guessing, a chance-corrected link is often closer to the scientific claim. Thus, for interaction claims, the key question is not only whether accuracy is binomial. It is whether the model defines additivity on the scale required by the task and by the claim (Loftus, 1978; McCabe et al., 2022; Wichmann & Hill, 2001).

Sum Scores as Bounded and Discrete Outcomes

Sum scores often combine a link-scale issue with a target-metric issue. They may be analyzed as Gaussian identity outcomes, transformed bounded scores, ordinal summaries, or item-level response models, depending on the scientific target and on the information available. We include them because many sum scores aggregate binary or ordinal response processes. For interaction claims, the risk is the same: the chosen metric or link defines the scale on which additivity is tested.

Sum scores make the link-function problem inseparable from measurement. Many psychological outcomes are reported as totals or averages across questionnaire items, task items, symptoms, errors, or ratings. Recent debates on sum scores make the same point from a psychometric angle: a total score can be useful and transparent, but it also encodes assumptions about how item responses represent the target construct (McNeish, 2023; McNeish & Wolf, 2020; Widaman & Revelle, 2023). These scores are often treated as if they were direct observations of an approximately continuous psychological dimension. But the latent construct and the observed score are different objects. The construct may be conceived as a continuous dimension, and may sometimes be reasonably approximated as smooth or roughly normal. The sum score is instead produced by a finite set of item responses through thresholds, endorsement probabilities, success probabilities, category labels, and scoring rules. It is therefore a remapped version of the target continuum, not the continuum itself.

This point matters even when the construct is otherwise well behaved. A normally distributed latent variable can still generate a bounded, discrete, skewed, or non-interval observed score. A total based on binary or ordinal items has only a limited number of possible values. It has a lower and upper bound. It can be skewed when items are too easy, too hard, rarely endorsed, or almost always endorsed. It can be compressed near floor or ceiling, but compression is not restricted to the endpoints. Adjacent raw-score values may correspond to different distances on the latent scale at different regions of the score range. These features are common in psychological variables (Micceri, 1989), and they are central to why treating ordinal responses as metric can produce false alarms, missed effects, reversals, and misleading interactions (Liddell & Kruschke, 2018).

Sum scores can be useful, transparent, reliable enough for many purposes, comparable across studies, and pragmatically tied to how instruments are scored in applied settings (Widaman & Revelle, 2023). A Gaussian identity model for a sum score still makes a scale claim. It assumes that effects are additive in raw-score units on the observed score scale. In an interaction model, the no-interaction baseline is that the expected raw-score effect of one predictor is constant across levels of the other predictor. Sometimes raw-score units are the target estimand. For example, a clinician, teacher, or researcher may care about a two-point difference on the reported score because decisions and interpretations are made on that metric.

Gaussian identity models are more defensible when the score has many possible values, the distribution is not strongly skewed, observations are not concentrated near floor or ceiling, residual patterns are not strongly tied to fitted values, and the substantive claim is explicitly about raw-score differences. They are less defensible when the score has few levels, is built from ordinal item responses with unclear spacing, shows floor or ceiling concentration, or is interpreted as evidence about moderation on a latent construct scale.

The problem becomes sharper when the verbal claim moves from the manifest score to the underlying construct. A group near the upper bound may show a smaller observed treatment effect not because the treatment is less effective on the construct, but because the score has less room to increase. A group located near a dense region of thresholds may show a larger raw-score change because the instrument is more sensitive there. The product term can then partly reflect the target metric rather than genuine moderation in the psychological process. A closely related problem has been shown in item-level work on genotype-by-environment interactions, where summed item scores can mask genuine interactions or generate spurious ones through scale properties of the observed score (Molenaar & Dolan, 2014). This is the sum-score version of outcome-model misspecification and target-metric mismatch (Domingue et al., 2024).

Several pragmatic alternatives can make the scale assumption more explicit. For a manifest sum score, a Gaussian identity model can be a defensible approximation when the claim is explicitly about raw-score units, the score has many levels, predicted values remain in range, and floor or ceiling compression is limited. When the claim is closer to an underlying construct, the better solution is an explicit measurement model, such as item-level ordinal, IRT-inspired, or SEM models. A bounded-score probit model can be used only as a sensitivity analysis on the manifest total. It is not a substitute for a measurement model. This point is also consistent with the sum-score debate: replacing a total score with a latent-score or item-level model changes the assumptions, rather than removing them (McNeish, 2023; McNeish & Wolf, 2020; Widaman & Revelle, 2023). The role of the alternatives considered here is not to establish the true scale, but to test whether the interaction conclusion depends on the manifest sum-score metric (Bürkner & Vuorre, 2019; Domingue et al., 2024; Liddell & Kruschke, 2018).

The recommendation is to make the assumed scale visible. Researchers can inspect the score distribution, plot predicted values near the bounds, report response-scale contrasts, and compare Gaussian identity results with bounded-score probit, ordinal, bounded-count, or item-level alternatives when these are plausible. Stable interactions across these specifications are easier to defend. Interactions that appear only on the manifest sum-score metric can still be informative, but they should be framed as properties of that score and interpreted as construct-level moderation only with supporting measurement assumptions.

Within-Family Link Choice

The link-function problem also arises within the same outcome family. For binary, binomial, ordinal, count, or positive continuous outcomes, the family can be broadly appropriate while the link remains underjustified. This is easy to miss because many default links are reasonable in many datasets. Logit and probit links, for example, often produce very similar fitted probabilities in the middle of the response range, especially after the usual approximate rescaling of coefficients. That practical similarity is real. It explains why many applied conclusions are unchanged by switching between the two links. However, it is not a general guarantee, especially when the claim depends on interaction, curvature, floor effects, ceiling effects, or tail behavior. Link misspecification can affect likelihood-based inference in binary GLMs and GLMMs (Czado & Santner, 1992; Yu & Huang, 2019).

The choice between logit and probit involves more than coefficient rescaling. A logit link is natural when the substantive interpretation is in odds or odds ratios. In that case, additivity on the linear predictor scale means additivity in log odds, and exponentiated coefficients have a direct odds-ratio interpretation. A probit link is natural when the response can be represented as threshold crossing on an approximately normal latent variable, such as ability, signal strength, response tendency, propensity, or latent correctness. The two links encode different stories about the scale on which effects combine.

For interaction claims, this distinction is more consequential than it may appear from main effects alone. Two links can give nearly overlapping fitted probabilities in the central range but imply different distances on the link scale near the floor, ceiling, chance level, or distributional tails. A product term in a logit model asks whether effects depart from additivity in log odds. A product term in a probit model asks whether they depart from additivity on a latent-normal threshold scale. These null hypotheses are often similar, but they are not identical. Moreover, when the substantive claim concerns observed probabilities, accuracy, endorsement, or risk, the product-term coefficient is not itself the response-scale interaction. The response-scale interaction must be stated as a contrast, marginal effect, or difference between predicted probabilities (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019).

This fitted-data example is deliberately simple. Logit and probit often lead to similar conclusions. Small differences in link curvature can be hard to see in the fitted probability curves, yet remain structured rather than random. If a moderation claim depends on the part of the predictor range where this discrepancy is largest, especially near a floor, ceiling, chance level, or tail, link choice can become inferentially consequential.

Logit-versus-probit sensitivity is therefore more than a cosmetic robustness check. A moderation claim that appears only under one plausible link, changes sign under another, or depends on predictions near a boundary should be reported as conditional on the chosen link. Stable response-scale conclusions across logit, probit, and other theoretically defensible links are easier to defend. Logit and probit often agree; their differences matter when the interaction claim depends on the baseline scale.

The same logic applies beyond the logit-probit pair. A binomial family specifies a sampling model for successes out of trials, but it does not decide whether additivity is most plausible on a logit, probit, complementary log-log, flexible generalized-logistic, or chance-corrected scale. Classical work on binary-response links already treated alternative links as alternative scales for additivity, including symmetric and asymmetric departures from the logistic link (Aranda-Ordaz, 1981; Stukel, 1988). In forced-choice tasks, the link question includes whether the lower asymptote should be 0, as in a standard binomial link, or the theoretical chance level of the task. An ordinal family does not decide which cumulative link is appropriate, and applied ordinal-regression work explicitly treats logit, probit, complementary log-log, negative log-log, and cauchit links as different modeling choices (Smith et al., 2020). A Gamma family does not decide whether the mean should be additive on the identity, log, or inverse scale. For interaction testing, these choices are not merely alternative parameterizations. They define different scales on which no interaction is assumed.

When link choice may matter less

Link choice will not change every substantive conclusion. In many applications, plausible links lead to nearly identical fitted values and similar response-scale summaries. The risk is smaller when predicted probabilities remain far from floor and ceiling, effects are modest, the relevant predictor range is well observed, and candidate links produce almost overlapping curves in that range. Crossover or qualitative interactions are also less vulnerable to monotone rescaling in the classical sense, so the present framework is most relevant for quantitative, non-crossover interaction claims (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012). As Figure 4 illustrates, the relevant issue is whether the pointwise discrepancy between two fitted curves overlaps with the region that drives the interaction claim. Link choice is also less threatening when the interaction is reported as a response-scale quantity, such as a marginal effect or contrast, and when that quantity is robust across theoretically defensible links (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019). Some theories concern average differences in a central range, away from saturation, guessing, floors, or ceilings; in those cases, link sensitivity may be small. Sensitivity analyses let readers see whether the interaction claim is tied to a narrow modeling convention or survives plausible alternatives.

Diagnostics Are Necessary but Not Sufficient

Diagnostics should be part of link-aware modeling, but they cannot settle the link-function question by themselves. The problem is not uniform diagnostic failure. It is diagnostic misalignment: a wrong link can generate a pseudo-interaction that becomes statistically detectable, while the checks used to diagnose the model may remain quiet, have limited power, or flag only broad misfit. In that situation, the interaction test can become sensitive to the wrong scale before the diagnostic evidence becomes specific enough to identify its source.

Pregibon-style link tests are useful because they target link adequacy more directly than generic residual plots (Pregibon, 1980). They expand the fitted GLM by adding terms derived from the fitted link, then ask whether the expanded model improves the fit. A significant added-term test suggests that the assumed link may be inadequate, but it does not identify the correct link. The signal may also reflect a missing nonlinear predictor, an omitted covariate, an omitted interaction, or another problem in the systematic component. A non-significant result is also limited. It means that this diagnostic, in this sample, did not detect this particular departure.

Simulation-based residual diagnostics provide a broader check. DHARMa implements randomized quantile residuals by simulating new responses from the fitted model and comparing the observed response with the fitted predictive distribution (Dunn & Smyth, 1996; Hartig, 2024). These residuals are easier to interpret across GLMs and GLMMs than raw, Pearson, or deviance residuals. In the diagnostic simulation below, we use four DHARMa checks: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual quantile patterns over the focal predictor. These checks can detect broad misfit, dispersion problems, and structured residual patterns. But they do not by themselves distinguish link misspecification from family misspecification, a missing nonlinearity, a missing interaction, or a target-metric mismatch.

Model comparison tools, such as AIC, BIC, WAIC, or LOO, add another source of evidence, but only within a clear candidate set. For link diagnostics, their cleanest use is to compare candidate links while holding fixed the observed outcome, likelihood family or support, and structural formula. In that use, AIC can compare Poisson-log and Poisson-identity fits to the same counts, binomial-logit and binomial-probit fits to the same successes and trials, or Gamma-log and Gamma-inverse fits to the same positive outcome. It can also compare a standard binomial link with a chance-corrected binomial link when both define likelihoods for the same binomial data. It is less direct when the comparison changes the outcome, the likelihood support, or the measurement metric, which is why we do not treat sum-score alternatives as a simple AIC contest among links. More importantly, information criteria compare predictive fit among the models supplied by the analyst. They do not decide which scale defines the scientific no-interaction claim. A wrong or incomplete link may even be favored when the product term absorbs part of the curvature created by the misspecification. Thus, a lower AIC is evidence about relative fit in the candidate set, not proof of link adequacy.

For interaction claims, diagnostics should therefore be used as part of the argument, not as a final answer. A useful order is: first identify constraints from the task, scoring rule, and measurement process; then define a small set of plausible outcome families, links, and target metrics; then use residual diagnostics, link checks, calibration, and information criteria as supporting evidence; finally, report whether the interaction conclusion is stable across the plausible specifications. A quiet diagnostic result should not end the assessment when the outcome has known constraints. For example, in a forced-choice task with a known chance level, a standard binomial logit can fit many aspects of the data while still encoding the wrong lower asymptote for the scientific claim. If the claim concerns response-scale moderation, researchers should report response-scale contrasts or marginal effects and examine whether those quantities are stable across plausible links (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). If the claim concerns additivity on a latent or linear-predictor scale, the chosen link needs a substantive justification.

Simulation Studies

The previous sections established the conceptual problem. The simulations ask how it translates into inferential risk when the true data-generating process is known. They are mechanism demonstrations, not prevalence estimates. Each scenario isolates a specific scale problem under controlled conditions: a chance floor, a bounded and discrete sum score, a within-family link choice, or a diagnostic comparison between the wrong-link interaction signal and model checks. Across simulations, the true product-term interaction is set to zero on the generating scale. We then fit models that impose different scales of additivity and record how often they detect a statistically significant interaction. The target is controlled quantification of link- or metric-induced pseudo-interactions rather than an estimate of how often such cases occur in the literature (Domingue et al., 2024; Rimpler et al., 2026).

The scenarios are illustrative mechanism demonstrations with a limited target. They were selected because they are common enough in psychological research, formally interpretable, and tied to constraints that can be stated before seeing the data. They also reflect a common interaction-testing setting in psychology: researchers begin from effects that are already known, such as group differences, treatment-control contrasts, developmental trends, and condition effects, and ask whether such effects are heterogeneous across a second predictor that also carries an established main effect. In this setting, two main effects jointly move observations across a curved link function. Curvature is therefore most able to create apparent heterogeneity when both main effects are present, which is the regime targeted here. The purpose is to show when a zero product term on one scale can become a detectable product term on another scale. They do not estimate prevalence, and they do not imply that the same false-positive rates should be expected for all bounded, discrete, ordinal, or positive outcomes.

Pseudo-interactions as power against the wrong null

A link-induced false-positive interaction is not only a sampling accident. Under the correctly specified GLM link, the no-interaction null states that effects are additive on the scale defined by that link. Under a wrong link, the same expected means may imply a nonzero interaction on the fitted link scale. A significant product term can therefore be understood as power to detect a pseudo-interaction created by the mismatch between the generating link and the fitted link.

This idea extends the classic scale-dependence argument for interactions (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012) and connects it to work on GLM and outcome-model misspecification (Domingue et al., 2024). It also complements work showing that product-term coefficients in nonlinear GLMs do not automatically equal interaction effects on the response scale (Ai & Norton, 2003; McCabe et al., 2022; Mize, 2019).

In a simple 2×2 design, let $\mu_{00}, \mu_{10}, \mu_{01}, \mu_{11}$ be the four expected response means generated with no product-term interaction on the target link scale. If the fitted model uses link g, the induced interaction on that fitted scale is

$$\Delta_g = g(\mu_{11}) - g(\mu_{10}) - g(\mu_{01}) + g(\mu_{00}).$$

When g is the target link, $\Delta_g = 0$. When g is not the target link, Δ_g can differ from zero even though the generating model contains no product term. This is the finite-difference version of the curvature argument introduced above. The risk depends on the size of this pseudo-effect relative to its standard error. This is why link misspecification is often small in central ranges where plausible links are nearly equivalent, but larger when the relevant comparisons involve floors, ceilings, chance levels, or distributional tails.

The box summarizes the logic behind the simulation estimand. The reported detection rates are misspecification-induced detection rates. They are not estimates of ordinary Type I error under a correctly specified GLM, and they are not prevalence estimates for the psychological literature.

The simulations quantify two quantities. First, for each substantive scenario, we estimate the probability of detecting an interaction when the generating model contains no product term on the relevant scale. Second, in the diagnostic simulation, we estimate how often residual checks, a Pregibon-style link check, and same-formula AIC comparisons across candidate links flag the same misspecification that produced the pseudo-interaction. This comparison matters because link-oriented tests, simulation-based residual diagnostics, and information criteria can be useful, but they do not guarantee that the fitted link is adequate for the interaction claim being tested (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980).

The repository reports the full data-generating processes, scripts, and extended output. The main text reports compact summaries.

Simulation 1: Forced-Choice Accuracy

Simulation 1 quantifies the forced-choice problem under controlled conditions. We generated binomial accuracy data with a .50 chance floor and no age-by-group product term on the chance-corrected logit scale. We then fit four models: a Gaussian identity model for proportions, standard binomial-logit and binomial-probit models, and a chance-corrected binomial-logit model. The outcome of interest was the false-positive rate for the age-by-group product term, that is, how often each fitted model detected an interaction that was absent on the generating scale.

Note. Successful fits are the number of replications in which the model converged and returned an interaction test. Estimates for models with failed fits are summarized over successful replications only.

The results match the mechanism described above. The chance-corrected model remains near nominal alpha across scenarios, because it matches the generating scale. The standard binomial and Gaussian identity models show elevated false-positive rates that are largest when one group's performance is close to the .50 floor, exactly where the misplaced lower asymptote creates the strongest curvature mismatch.

The chance-corrected link can be numerically demanding. When observed performance is close to the chance floor, the likelihood may contain little information about changes on the above-chance scale, and frequentist fits can become unstable or fail to converge. This instability is information about the design and the data: the task may provide limited evidence about individual or group differences above guessing in that region. In applied work, analysts should report convergence problems, inspect predicted probabilities near the floor, and treat the chance-corrected fit as part of a small sensitivity analysis over plausible links. Bayesian implementations with weakly informative priors, or item-level psychometric or IRT-inspired models, may stabilize estimation in some settings, but they also add assumptions that should be stated and checked.

The Gaussian identity model and the standard binomial links can show elevated false-positive rates, especially when the scenario places one group closer to the chance floor. Standard binomial models do not always produce false interactions. When the theoretical floor is .50 but the fitted link treats 0 as the lower asymptote, however, the product term may become a way to repair a wrong no-interaction baseline.

Simulation 2: Sum Scores

Simulation 2 makes the measurement-scale problem concrete. We generated data on a latent continuous scale with a continuous predictor, a group effect, and no true x-by-group product term. We then converted the latent variable into ordinal item responses, with each item scored from 0 to 3, and summed the items. This setup represents a common psychometric scoring situation in which a bounded manifest score summarizes an underlying response process. The interaction was therefore absent on the latent generating scale, but the observed score was bounded, discrete, and differentially compressed near the floor and ceiling.

The simulation compares three analyses, but only to separate the relevant scales: Observed sum-score identity, Gaussian-probit bounded score, and Latent generating scale. The observed sum-score identity model is the familiar Gaussian identity analysis of the manifest total. The Gaussian-probit bounded-score model treats the rescaled total as a bounded outcome and asks whether changing the scale of additivity changes the interaction conclusion. The latent generating scale is an idealized benchmark available only in the simulation, because the true latent theta is known. It is included to show what the interaction test would target if the generating construct scale were directly observed.

The Gaussian-probit bounded-score model is included as a sensitivity model, not as a recommended measurement model. It fits a continuity-corrected score proportion with a Gaussian likelihood and a probit link, then rescales predictions to the sum-score metric. The probit link is more interpretable here than a logit link because its linear predictor is on a standard-normal, z-like scale. In the idealized threshold-normal setting used in the simulation, this makes the coefficient scale closer to standardized latent differences. This interpretation remains conditional on the measurement assumptions and should not be read as a literal Cohen's d for the target construct.

The results are conditional on the region of the score range. When the manifest score is approximately linear over the observed range, far from its bounds, and the claim is about raw-score units, the Gaussian identity analysis closely approximates the latent-scale result. When the score compresses the latent scale unevenly, the raw-score no-interaction baseline can diverge from the latent no-interaction baseline. A bounded-score probit model can be useful as a sensitivity analysis because it asks whether the conclusion changes when the expected score proportion is mapped through a standard-normal link. It is still a model of the manifest total, not a measurement model for the construct. When the claim concerns an interaction between latent constructs, the more direct solution is to model the measurement structure explicitly, for example with SEM, item-level ordinal models, IRT-inspired models, or other explicit measurement models. These models still require assumptions about thresholds, links, loadings, and latent-response metrics, so they do not remove the scale problem. They make the measurement part of the interaction claim explicit.

Simulation 3: Within-Family Link Choice

Simulation 3 moves from the fitted-data illustration in Figure 4 to a repeated-sampling question: can within-family link choice inflate false-positive interaction rates when the outcome family, experimental design, and random-effect structure are held fixed? We generated repeated binary responses in a two-group, two-condition design, with 15 trials per participant-condition cell and a subject random intercept. The true condition-by-group product term was zero on the generating link scale. We then crossed the generating link and fitted link, using logit and probit in both roles, and fitted all models as random-intercept GLMMs.

The only misspecification in this simulation is within-family link choice: the outcome family, the experimental design, and the random-effect structure are held fixed. Under the wrong fitted link, the same cell probabilities imply a small but deterministic nonzero product term on the fitted link scale, which becomes detectable with enough information.

Simulation 3 is a boundary-sensitive stress test. Logit and probit links often lead to similar conclusions, especially when probabilities are concentrated in the middle of the response range. That practical similarity is real, and it explains why many applied analyses are unaffected by switching between the two links. The contribution of this simulation is to show that the link problem extends beyond obvious misspecification, such as treating a .50 floor as 0 or using a continuous model for binary data. It can appear even among links that share the same family, the same bounds, and nearly overlapping fitted curves in the middle of the probability range. Relative to the forced-choice case, the observed false-positive rates are smaller. In inferential terms, they remain large: about 13% to 15% compared with the 5% nominal target, or roughly a tripling of the nominal error rate in a setting many applied researchers would treat as harmless. Simulation 3 quantifies this risk, and Figure 4 shows why visually similar fitted curves do not guarantee interaction robustness.

The table reports false-positive rates relative to the generating-link null. The median product-term coefficient and its standard error are on the fitted link scale, so they should not be read as response-scale changes in probability. Figure 7 makes the Monte Carlo result explicit: even with the same binary outcome family, logit and probit define slightly different no-interaction baselines, and this difference can become inferentially visible in repeated samples. A result that is stable across both links is easier to defend. A result that appears only under one link should be reported as link-conditional, or translated into a response-scale estimand and checked across plausible links.

Diagnostic Simulation

Finally, we examined whether common diagnostics would flag the same link problems that produced pseudo-interactions. This simulation is not a general benchmark of all possible checks. It asks whether diagnostic evidence appears before, after, or at the same time as the false interaction.

We used five scenarios, each tied to a previous example or matched counterpart in the paper. The first is the simple count example in Figure 3. Data are generated from a Poisson-log model in which expected errors decline with age, with no age-by-group product term on the log-count scale. The coefficients are the same as in the figure: intercept 2.60, age effect -0.55 per year after age 6, and group effect 0.65. The fitted model keeps the Poisson family and the interaction formula, but uses an identity link. This makes the comparison a same-family link comparison, not a comparison between Poisson and Gaussian models.

The second scenario is the lower-performance forced-choice scenario from Simulation 1. Data are generated with a .50 chance floor, 20 trials per participant, intercept -0.80, age effect 0.60, group effect -0.90, and no age-by-group product term on the chance-corrected logit scale. The fitted model is a standard binomial-logit interaction model, which respects the binomial trial structure but places the lower asymptote at 0 rather than at chance.

The third scenario mirrors the probit-generated, logit-fitted case in Simulation 3. Data are repeated binary trials with a subject random intercept, 15 trials per condition cell, and latent ICC .30. The fixed effects are the probit reference values used in Simulation 3: intercept 1.50, group effect -1.00, condition effect -1.00, and no group-by-condition product term. The fitted model is the corresponding random-intercept logit GLMM with the same interaction formula.

The fourth scenario is a continuous-predictor counterpart to the previous binary repeated-trials case. It keeps the probit data-generating link, the logit fitted link, the group effect, the focal-predictor effect, the subject random intercept, and the latent ICC, but replaces the two-level condition with a continuous predictor sampled from 0 to 1. This scenario asks whether the same logit-versus-probit issue is visible when the focal predictor has enough unique values for continuous-predictor residual checks to be better identified.

The fifth scenario connects to the Gamma example in Figure 1. Data are generated from a Gamma model with a log link, shape 8, baseline mean 400 when x = 0 and group = 0, x effect 0.25 on the log-mean scale, group effect 0.25, and no x-by-group product term. The fitted model keeps the Gamma family and the interaction formula, but uses the inverse link.

In each replication, diagnostics were applied to the misspecified interaction model, because this is the model an analyst would usually inspect after finding the interaction. This design targets a realistic post-estimation question: does the same fitted model that produced the product-term finding also show enough diagnostic evidence to warn the analyst about the link problem? The choice is demanding because the product term can absorb part of the curvature created by the wrong link. A diagnostic workflow applied to an additive model before adding the product term would answer a different question. The present simulation asks whether ordinary checks protect the interaction conclusion once the misspecified interaction model has been fitted.

The residual workflow used four DHARMa checks: residual uniformity, dispersion, residual quantile patterns over fitted values, and residual quantile patterns over the focal predictor (Dunn & Smyth, 1996; Hartig, 2024). We also used a Pregibon-style added-term link check as a more link-oriented diagnostic (Pregibon, 1980).

Finally, we used AIC only for relative comparisons among prespecified candidate mean models fitted to the same response data with the same interaction formula. In the count and Gamma scenarios, the response family is held fixed and only the link changes. In the binary scenarios, the binomial sampling model is held fixed. The chance-floor scenario compares two binomial mean mappings: a standard logit model and a chance-corrected logit model with a design-implied lower asymptote. Thus, AIC varies the candidate mean-link mapping, not the presence or absence of the product term. We report the proportion of replications in which the lower AIC favored the target model. Because no inconclusive ΔAIC category is used, the complement favors the misspecified model when both AIC values are available. These values should be read as relative fit within a limited candidate set, not as evidence that AIC can identify the scientifically correct scale of additivity.

Table 7 documents diagnostic misalignment: the same misspecification that generates a pseudo-interaction often does not produce a clear diagnostic signal. In several same-family scenarios, AIC or the Pregibon-style check favored the target model, but protection was uneven and depended on which candidate models were compared. The starkest result is the chance-floor scenario: a psychologically common link problem produced a large pseudo-interaction rate while all four DHARMa checks remained near nominal levels and AIC favored the misspecified model.

Practical Starting Points for Link-Aware Interaction Testing

The applied output of the framework is a small set of outcome-specific starting points. These starting points are prompts for the choices that should be visible in a report: what family respects the observed response, what link defines the no-interaction baseline, what target metric answers the substantive question, and what sensitivity check is feasible for the design. Table 8 summarizes these choices for common psychological outcomes.

The link choice and the robustness check answer different questions. The preferred link should be justified from theory, task structure, or measurement whenever possible. When these considerations do not uniquely determine the link, probing the conclusion across plausible links is a transparency step: it shows which part of the conclusion depends on modeling conventions and which part does not. Stable conclusions are stronger; unstable conclusions can still be informative, but should be reported as link-conditional.

These entries are starting points; theory, design, and data quality still decide the final model, including whether the design is informative enough for the interaction claim (Baranger et al., 2023; Castillo et al., 2026). A link-scale coefficient can be useful when the scientific claim concerns that model scale. A response-scale contrast or marginal effect is often closer to the claim made in words (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). Marginal effects clarify the estimand, but they remain conditional on the fitted model and on the point or distribution at which they are evaluated.

Diagnostics, predictive checks, and information criteria can warn about misspecification or relative fit, but they provide secondary evidence for an interaction claim (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980). The central robustness question is whether the target estimand remains substantively stable across a small, theory-limited set of plausible links or measurement models. This is related to the logic of multiverse analysis, although the recommendation here is narrower: a targeted sensitivity analysis defined by the outcome, the task, and the scientific estimand. Stable conclusions across plausible specifications are easier to defend. Unstable conclusions should be reported as conditional on the modeling choice. This extends a basic reporting principle: choices that affect the claim should be visible, justified, and interpretable (Mustillo et al., 2018).

Discussion

Whether an interaction claim survives plausible link alternatives is a question that existing reporting practice rarely answers. Some claims will be stable across reasonable links, scales, and response-scale summaries; others are conditional on a modeling choice that is currently left implicit. The scale problem is well known (Cox, 1984; Loftus, 1978; Wagenmakers et al., 2012). In GLMs, it takes the practical form developed above. The same issue can also extend beyond interactions to nonlinear effects, because many psychological analyses search for curvature, thresholds, and changing slopes without first specifying the scale on which these patterns are expected (Domingue et al., 2024; McCabe et al., 2022; Rohrer & Arslan, 2021).

The preregistered review showed why this matters for current practice. Interaction tests were common, link functions were often implicit, and many interaction-testing articles included identity-link analyses with outcomes for which observed-scale additivity was not self-evident. The review documents this as a reporting gap: readers cannot evaluate which scale of additivity was assumed from the information provided.

The three worked examples show how family, link, and target metric can each carry the scale assumption behind an interaction test. Forced-choice accuracy is the clearest case: the binomial family is correct, but a standard link misplaces the lower asymptote. The logit-probit comparison shows that the problem persists even within a family, between links that look similar in the central range. Sum scores extend the problem to the target-metric level, where the manifest score is already a measurement mapping from item responses. One implication that goes beyond the specific examples is that target-metric problems can overlap with measurement non-invariance: when the mapping from latent response tendency to observed score differs across groups, conditions, or cultures, group differences in floor or ceiling exposure produce apparent interactions that reflect measurement rather than moderation. Ordinary floor and ceiling effects are therefore the visible extreme of a broader remapping problem that applies whenever the observed metric is not the theoretical metric.

The simulations were designed to quantify pseudo-interactions, not to estimate the prevalence of false interactions in the literature. They complement work showing that interaction estimates are often difficult to estimate precisely even when the model form is appropriate (Baranger et al., 2023; Castillo et al., 2026). They show how link and metric choices can generate significant product terms even when the true interaction is zero on the generating scale. The diagnostic simulation adds a practical caution: residual checks, Pregibon-style tests, and information criteria can be informative, but their protection was uneven and depended on the candidate set. The chance-floor scenario is the clearest psychologically relevant failure case: a wrong link generated a large pseudo-interaction while residual checks remained quiet and AIC favored the simpler standard link. This is diagnostic misalignment, not uniform diagnostic failure. Diagnostics should therefore be treated as evidence about model fit within a candidate set, not as proof that an interaction claim is robust to link choice (Dunn & Smyth, 1996; Hartig, 2024; Pregibon, 1980).

Five limitations bound the scope of the conclusions. First, the review was descriptive and targeted to five reference journals across major areas of psychology. The numerical estimates should therefore be read as estimates of reporting practice in high-standard outlets during the sampled year, not as a census of all psychology journals or all publication years. Second, we did not code whether published interactions were removable or nonremovable, because that distinction often cannot be recovered from published reports alone. Third, the simulations are mechanism demonstrations targeted at a specific regime: two established main effects that jointly expose observations to link-function curvature. They isolate scale problems under known data-generating processes and do not estimate prevalence or cover all realistic complications, such as item heterogeneity, lapses, response bias, nonlinear predictors, and missing data. Fourth, chance-corrected links are appropriate only when a chance floor is theoretically justified. They are not a universal replacement for standard binomial models. In forced-choice tasks, the chance floor may be fixed by design, estimated under stronger assumptions, or varied in sensitivity analyses. Finally, logit and probit links often lead to very similar conclusions, especially away from boundaries; our claim is that link choice can matter for interaction claims, not that it always does.

The main contribution is practical. Link-aware interaction testing gives researchers a common language for four judgments that are often separated in practice: what outcome family is plausible, what link defines the no-interaction baseline, what response-scale quantity answers the substantive question, and whether the conclusion survives plausible alternatives. This workflow complements existing guidance on nonlinear interactions, marginal effects, and prediction-based interpretation (McCabe et al., 2022; Mize, 2019; Rohrer & Arel-Bundock, 2026). Stable conclusions across the specifications in Table 8 strengthen the interaction claim. Unstable conclusions remain informative when reported as conditional on the modeling choice. The contribution is therefore operational: in GLMs, choosing the link is choosing the scale of the no-interaction claim.

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Table 1

Descriptive summary of the preregistered review. Formally incorrect identity-link analyses of constrained observed outcomes are cases where the reported outcome had formal constraints that make identity-link additivity formally incorrect as a generative model unless explicitly justified as an approximation or as the target observed-score metric.

Quantity	n	Denom.	%	95% CI
Eligible empirical articles	331	331	100.0%	[98.9%, 100.0%]
Eligible empirical articles testing at least one interaction	200	331	60.4%	[55.1%, 65.5%]
Interaction-testing articles using at least one non-identity link	58	200	29.0%	[23.2%, 35.6%]
Interaction-testing articles explicitly reporting the link function	46	200	23.0%	[17.7%, 29.3%]
Interaction-testing articles with formally incorrect identity-link analyses of constrained observed outcomes	144	200	72.0%	[65.4%, 77.8%]
Formally incorrect identity-link cases with at least one significant interaction	124	144	86.1%	[79.5%, 90.8%]
Eligible empirical articles not testing interactions	131	331	39.6%	[34.5%, 44.9%]

Table 2

Broad response-variable categories in interaction-testing articles. Categories are non-exclusive, so one article can contribute to more than one row.

Outcome type	n	Denominator	Percent
Binary / accuracy / proportions	84	200	42.0%
Sum scores / composites	66	200	33.0%
Ordinal / Likert / ratings	36	200	18.0%
Response times / durations	30	200	15.0%
Counts / error counts	9	200	4.5%
Neural / physiological measures	23	200	11.5%
Correlations / associations	8	200	4.0%
Difference / distance scores	17	200	8.5%

Table 3

Three analytic choices that determine the scale of an interaction claim. The choices are conceptually distinct, but they can overlap in applied analyses.

Analytic choice	Key question	Psychological example	Risk for interaction claims
Outcome family	Does the model respect the support and sampling process of the observed outcome?	Counts, proportions, or trial accuracies modeled as unbounded Gaussian observations.	Constraints, discreteness, or the mean-variance relation may be absorbed by a product term.
Link / additivity scale	Given a plausible family, on which scale should effects add if there is no interaction?	Forced-choice accuracy modeled with a standard binomial logit, although the task has a known chance floor such as .50.	The model may test additivity on the wrong scale, so a pseudo-interaction can arise even when the target link-scale model has only main effects.
Target metric / measurement mapping	Is the observed score the scale on which the theoretical interaction claim is meant to hold?	A Likert item or sum score treated as if raw-score units directly represented units of the target construct.	A product term may reflect thresholds, bounds, or scale compression rather than moderation of the underlying construct.

Table 4

Simulation 1: false-positive interaction rates and median model-implied observed-scale changes in the group gap for forced-choice accuracy data. CI = Wilson 95% confidence interval.

Scenario	Model	Fits	Sig	False-positive rate	95% CI	Median gap change
Lower performance	Gaussian identity	300	137	0.457	[0.401, 0.513]	-1.779
Lower performance	Standard binomial logit	300	191	0.637	[0.581, 0.689]	-1.769
Lower performance	Standard binomial probit	300	183	0.610	[0.554, 0.663]	-1.781
Lower performance	Chance-corrected binomial logit	178	14	0.079	[0.047, 0.128]	-1.225
Middle performance	Gaussian identity	300	40	0.133	[0.099, 0.176]	-0.865
Middle performance	Standard binomial logit	300	169	0.563	[0.507, 0.618]	-0.862
Middle performance	Standard binomial probit	300	148	0.493	[0.437, 0.550]	-0.866
Middle performance	Chance-corrected binomial logit	300	18	0.060	[0.038, 0.093]	-0.707
Higher performance	Gaussian identity	300	36	0.120	[0.088, 0.162]	0.654
Higher performance	Standard binomial logit	300	103	0.343	[0.292, 0.399]	0.555
Higher performance	Standard binomial probit	300	59	0.197	[0.156, 0.245]	0.617
Higher performance	Chance-corrected binomial logit	300	14	0.047	[0.028, 0.077]	0.526

Table 5

Simulation 2: false-positive interaction rates and median model-implied observed-scale changes in the group gap for sum-score data. Data were generated with no x-by-group product term on the latent scale. The latent generating scale is an idealized benchmark available only because theta is known in the simulation. CI = Wilson 95% confidence interval.

Scenario	Model	Scale	Fits	Sig	False-positive rate	95% CI	Median gap change
Middle of scale	Gaussian-probit bounded score	sum score	300	14	0.047	[0.028, 0.077]	-0.568
Middle of scale	Latent generating scale	latent theta	300	16	0.053	[0.033, 0.085]	0.018
Middle of scale	Observed sum-score identity	sum score	300	29	0.097	[0.068, 0.135]	-0.547
Near ceiling	Gaussian-probit bounded score	sum score	300	15	0.050	[0.031, 0.081]	1.413
Near ceiling	Latent generating scale	latent theta	300	16	0.053	[0.033, 0.085]	0.003
Near ceiling	Observed sum-score identity	sum score	300	112	0.373	[0.321, 0.429]	1.328
Near floor	Gaussian-probit bounded score	sum score	300	26	0.087	[0.060, 0.124]	-1.847
Near floor	Latent generating scale	latent theta	300	18	0.060	[0.038, 0.093]	0.011
Near floor	Observed sum-score identity	sum score	300	216	0.720	[0.667, 0.768]	-1.762

Table 6

Simulation 3: false-positive interaction rates for the condition-by-group product term in repeated-trial random-intercept GLMMs. CI = Wilson 95% confidence interval.

Generating link	Fitted link	Link status	Fits	Sig	False-positive rate	95% CI	Median beta interaction	Median SE
logit	logit	Matched link	300	16	0.053	[0.033, 0.085]	-0.001	0.075
logit	probit	Wrong link	300	45	0.150	[0.114, 0.195]	-0.041	0.042
probit	logit	Wrong link	300	41	0.137	[0.102, 0.180]	0.071	0.076
probit	probit	Matched link	300	15	0.050	[0.031, 0.081]	-0.003	0.043

Table 7

Diagnostic simulation across five link-misspecification scenarios. FP interaction reports how often the misspecified interaction model detected a product term absent on the generating link scale. DHARMa gives the range across prespecified residual checks, not an omnibus diagnostic. Pregibon reports the added-term link check. AIC favors target reports the proportion of replications in which the lower AIC favored the target interaction model over the misspecified interaction model, with the structural formula held fixed. The complementary proportion favors the misspecified model when both AIC values are available. CI = Wilson 95% confidence interval.

Case	Target model	Misspecified model	FP interaction	DHARMa	Pregibon	AIC favors target
Count errors	Poisson log	Poisson identity	0.942 [0.908, 0.964]	0.007-0.906 (4)	0.996	0.996
Chance-floor accuracy	chance-corrected binomial logit	standard binomial logit	0.620 [0.564, 0.673]	0.000-0.057 (4)	0.293	0.000
Binary repeated trials	binomial probit GLMM	binomial logit GLMM	0.090 [0.063, 0.128]	0.000-0.013 (3)	1.000	0.943
Binary continuous predictor	binomial probit GLMM	binomial logit GLMM	0.060 [0.038, 0.093]	0.000-0.007 (4)	1.000	0.947
Gamma mean response time	Gamma log	Gamma inverse	0.287 [0.238, 0.340]	0.000-0.060 (4)	0.350	0.787

Table 8

Practical starting points for link-aware interaction testing across common psychological outcomes.

Outcome type	Plausible starting point	Minimal sensitivity check
Forced-choice accuracy	Binomial logit/probit; chance-corrected link when the task floor is part of the estimand.	Compare standard logit/probit and chance-corrected links; report convergence and response-scale contrasts.
Binary/proportion outcome	Binomial logit/probit; consider cloglog for asymmetric processes.	Compare logit/probit; inspect calibration and contrasts near bounds or rare-event regions.
Likert/ordinal item	Ordinal logit/probit; metric model only with a clear observed-scale target.	Compare ordinal and metric summaries; report category probabilities or response-scale contrasts.
Sum score/composite	Gaussian identity when many levels stay far from bounds; otherwise consider item-level, ordinal, or bounded-score models.	Inspect bounds, skew, and score concentration; compare with item-level or bounded-score sensitivity models when feasible.
Counts/error counts	Poisson or negative binomial, usually with log link; identity only with an observed-count target.	Check overdispersion and zero inflation; compare count contrasts across plausible count models.
Response time/skewed positive outcome	Lognormal, Gamma log/inverse, or transformed Gaussian, matched to absolute, proportional, reciprocal, or transformed change.	Compare raw-, log-, and response-scale summaries; inspect tail fit and predicted values.

Figure 1

Same outcome family, different link functions. The grey construction lines are drawn from a target link: the chance-corrected logit in Panel A and the log link in Panel B. Vertical grey lines mark equal intervals on the target-link linear predictor scale. Horizontal grey lines show where those equally spaced target-link values fall on the observed response scale. The key point is that equal intervals on one link scale do not remain equal intervals under another link. Thus, models with the same outcome family can imply different no-interaction baselines, even when their fitted curves look similar over the observed range.

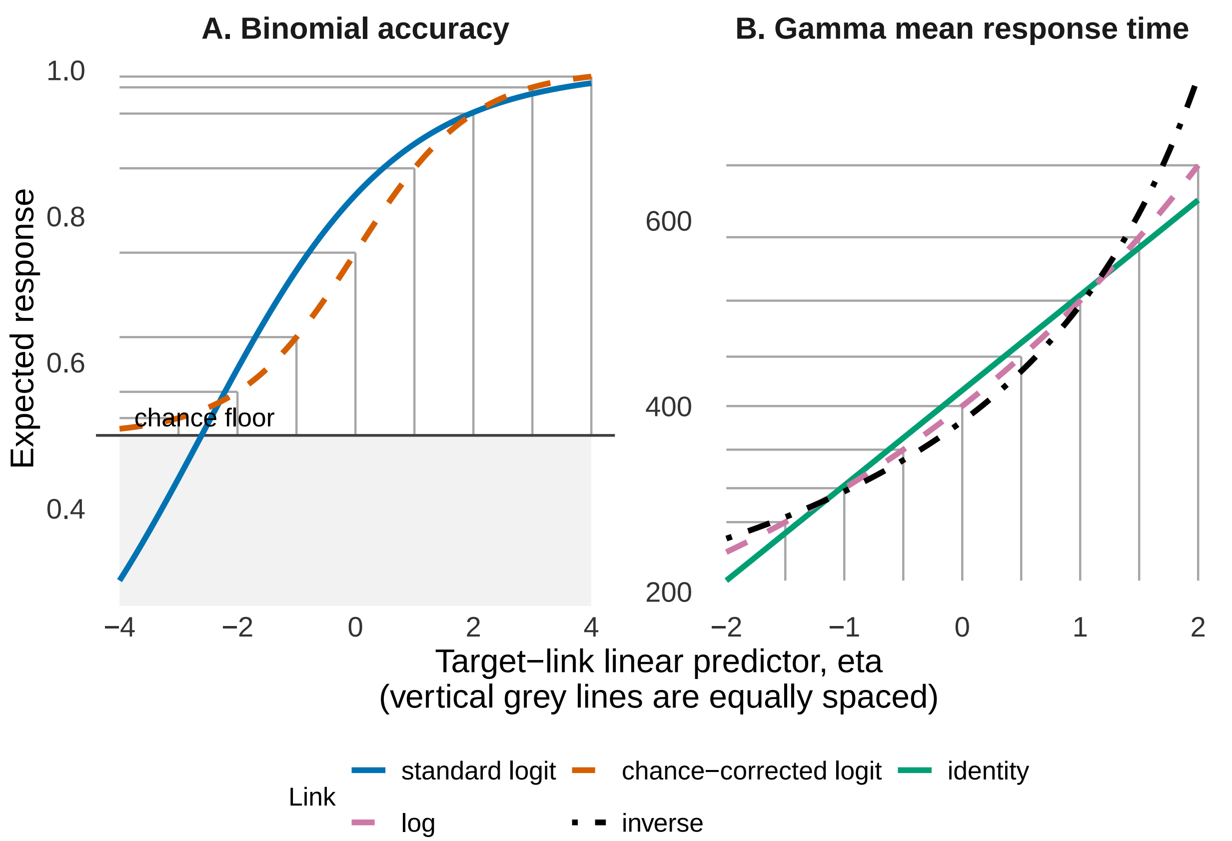

Figure 2

Same data, different link functions. Data were simulated from a 2-alternative forced-choice accuracy task with 50 trials per participant and a 0.50 chance floor. The data-generating model included age and group effects, but no age-by-group interaction on the chance-corrected 2-AFC logit scale. Panel A shows the true response-scale pattern implied by that model. Panel B shows one binned simulated dataset fitted with three additive models: a Gaussian identity model, a standard binomial logit model, and a chance-corrected binomial link. The Gaussian identity model is simple and interpretable on the observed accuracy scale, but it can produce predictions outside the 0-to-1 range and ignores the binomial sampling process. The standard binomial logit model respects the binomial process and the 0-to-1 range, but assumes a lower asymptote at 0. The chance-corrected binomial link incorporates the task's chance floor. The question is which scale defines the scientific null hypothesis of no moderation.

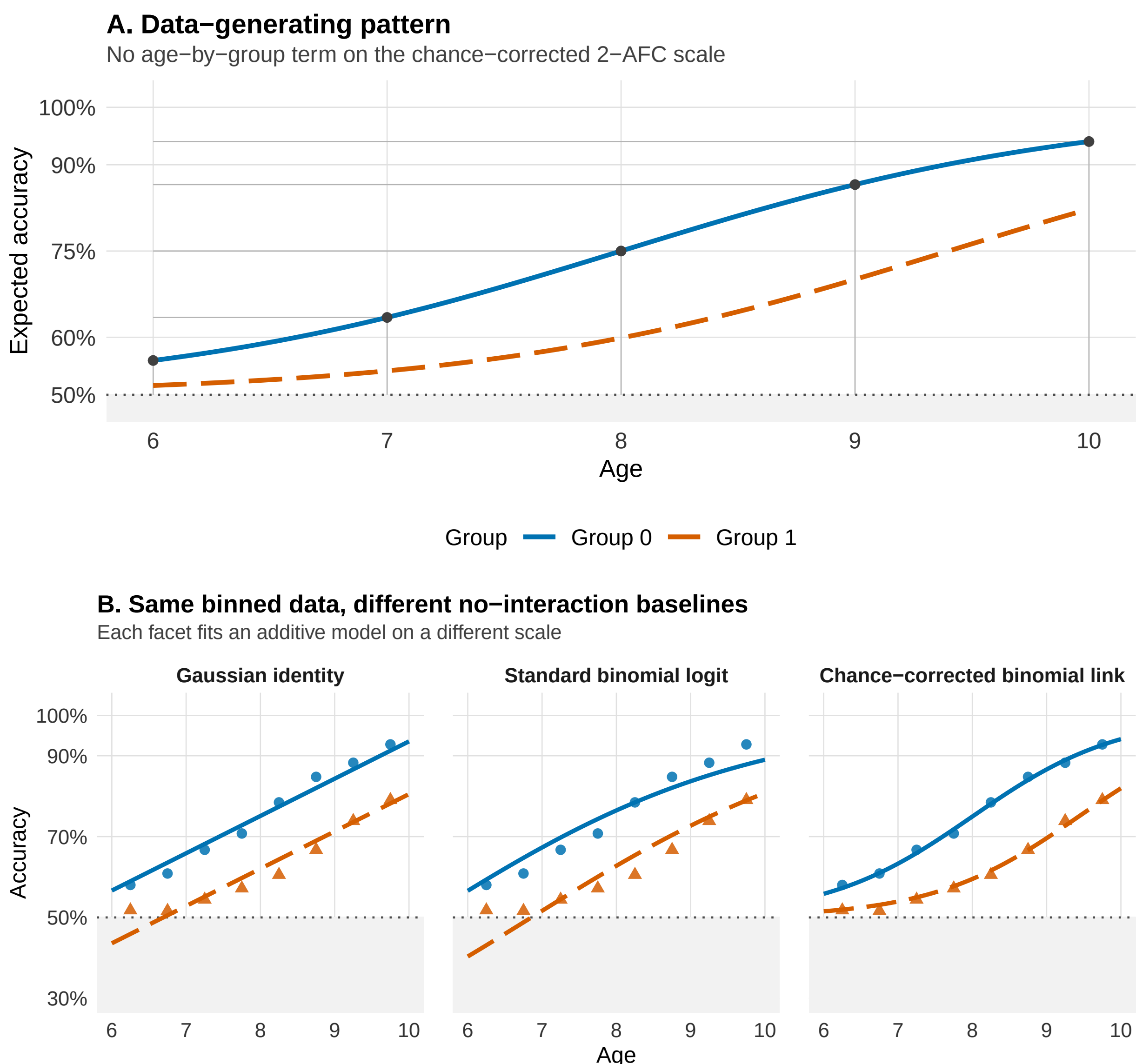

Figure 3

A simple count example. The two groups/conditions have the same age effect on the log-count scale and no age-by-condition product term on that scale, but after exponentiating the linear predictor, expected error counts decrease nonlinearly on the observed count scale, and differences between conditions appear to change with age.

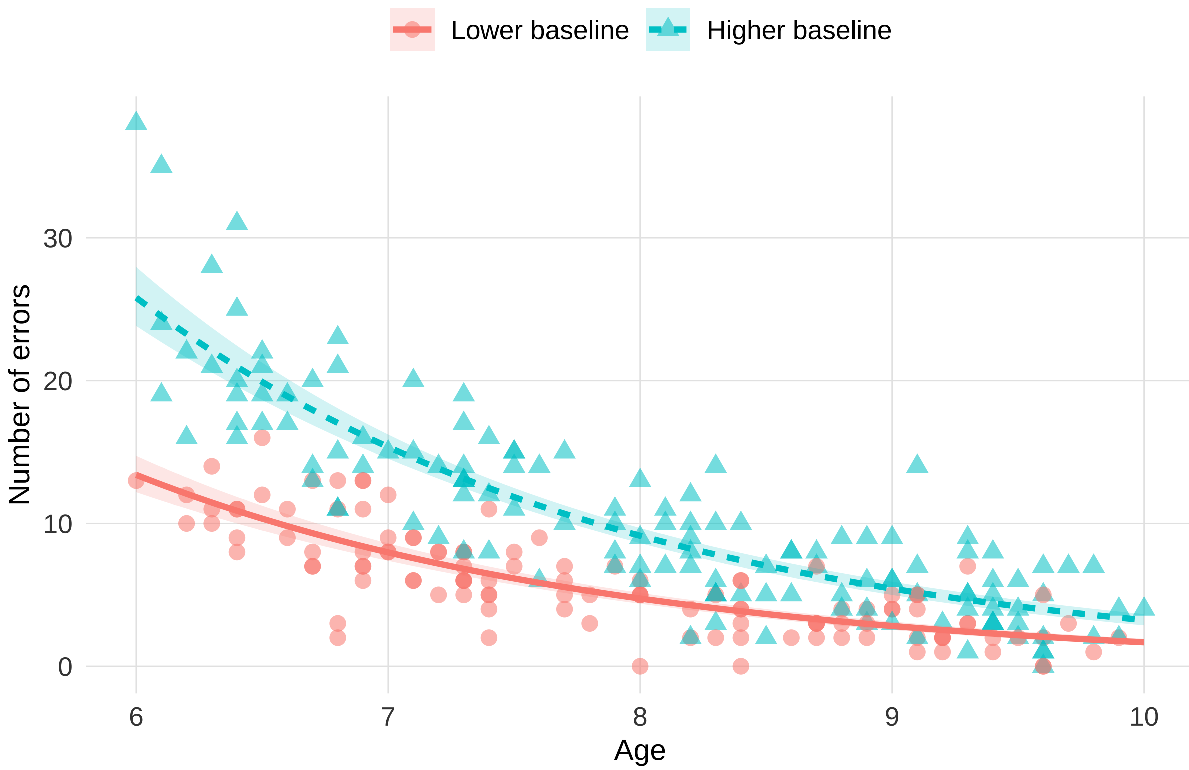

Figure 4

Fitted logit and probit models on one simulated binary dataset. The data were generated from a logistic curve, and both a binomial-logit and a binomial-probit GLM were fitted to the same responses. Panel A shows binned observed proportions together with the two fitted probability curves. Panel B shows the pointwise difference between the fitted probabilities, computed as logit minus probit. The curves can look nearly interchangeable while still implying a structured discrepancy across the predictor range. For interaction claims, this matters when the relevant comparison depends on regions where the two fitted links allocate different response-scale room for an effect.

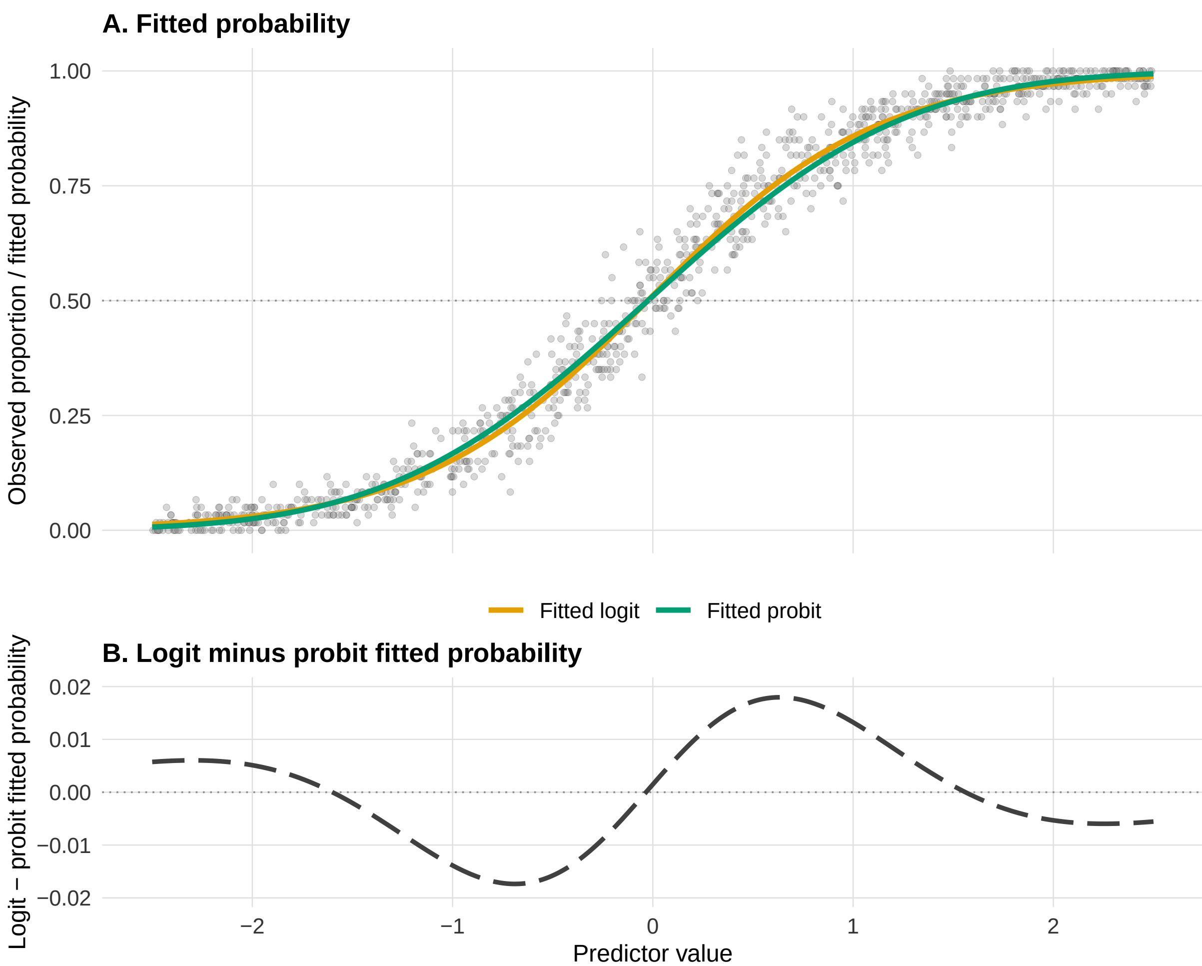

Figure 5

Simulation 1: forced-choice accuracy with a task-implied chance floor. Panel A shows the deterministic scenario curves generated by the chance-corrected logit model. The grey region marks values below the .50 chance floor, and the dashed horizontal line marks chance performance. Panel B shows the false-positive rate for the age-by-group product term across repeated simulations. Data are generated with no age-by-group interaction on the chance-corrected scale.

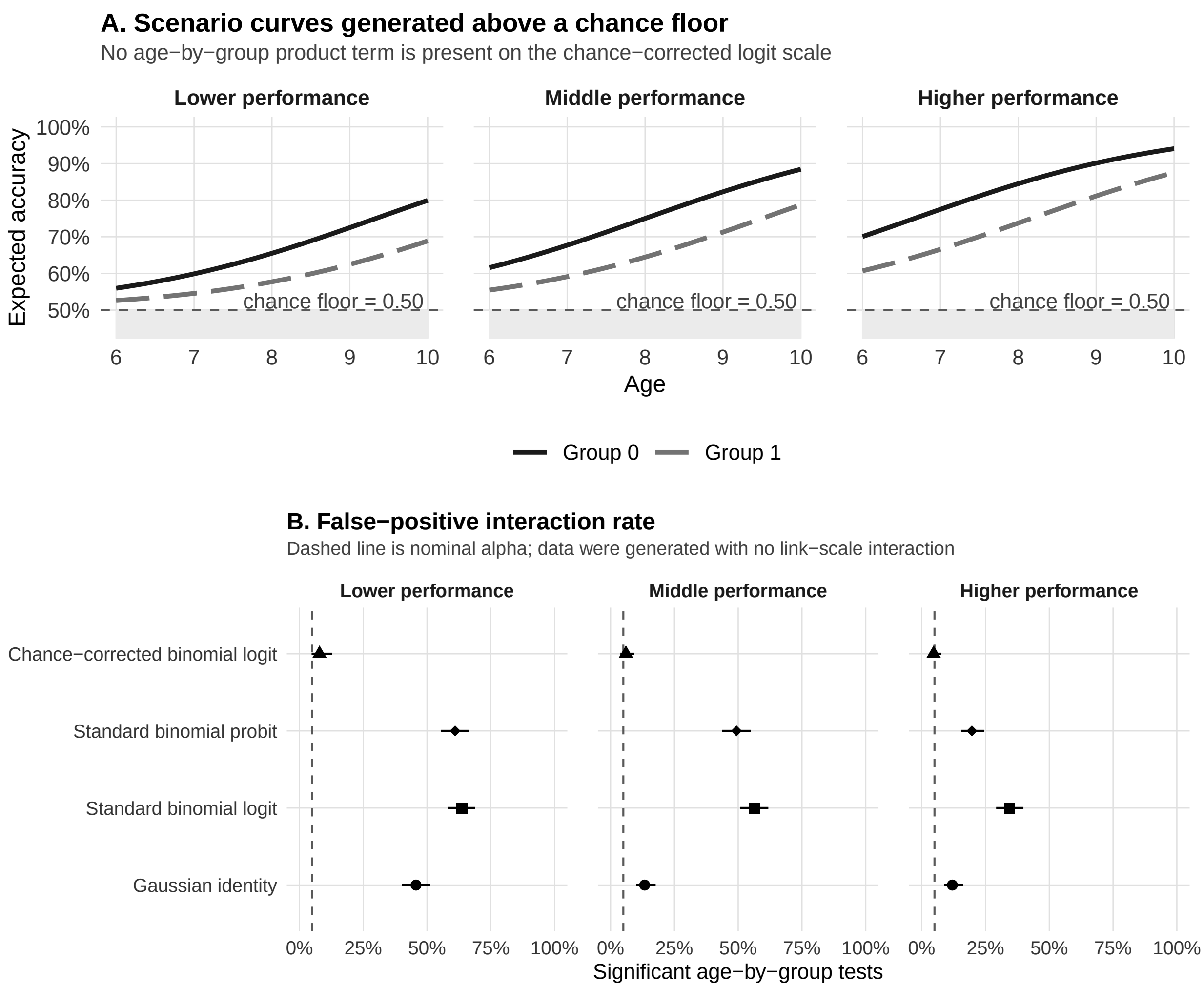

Figure 6

Simulation 2: sum scores as bounded and discrete outcomes. Panel A shows how a latent additive data-generating model, with no x-by-group product term, is mapped onto the observed sum-score scale under middle, floor, and ceiling scenarios. Panel B shows false-positive interaction rates across repeated simulations. The latent generating scale is an idealized benchmark available only because theta is known in the simulation, not a model available in ordinary applied data analysis.

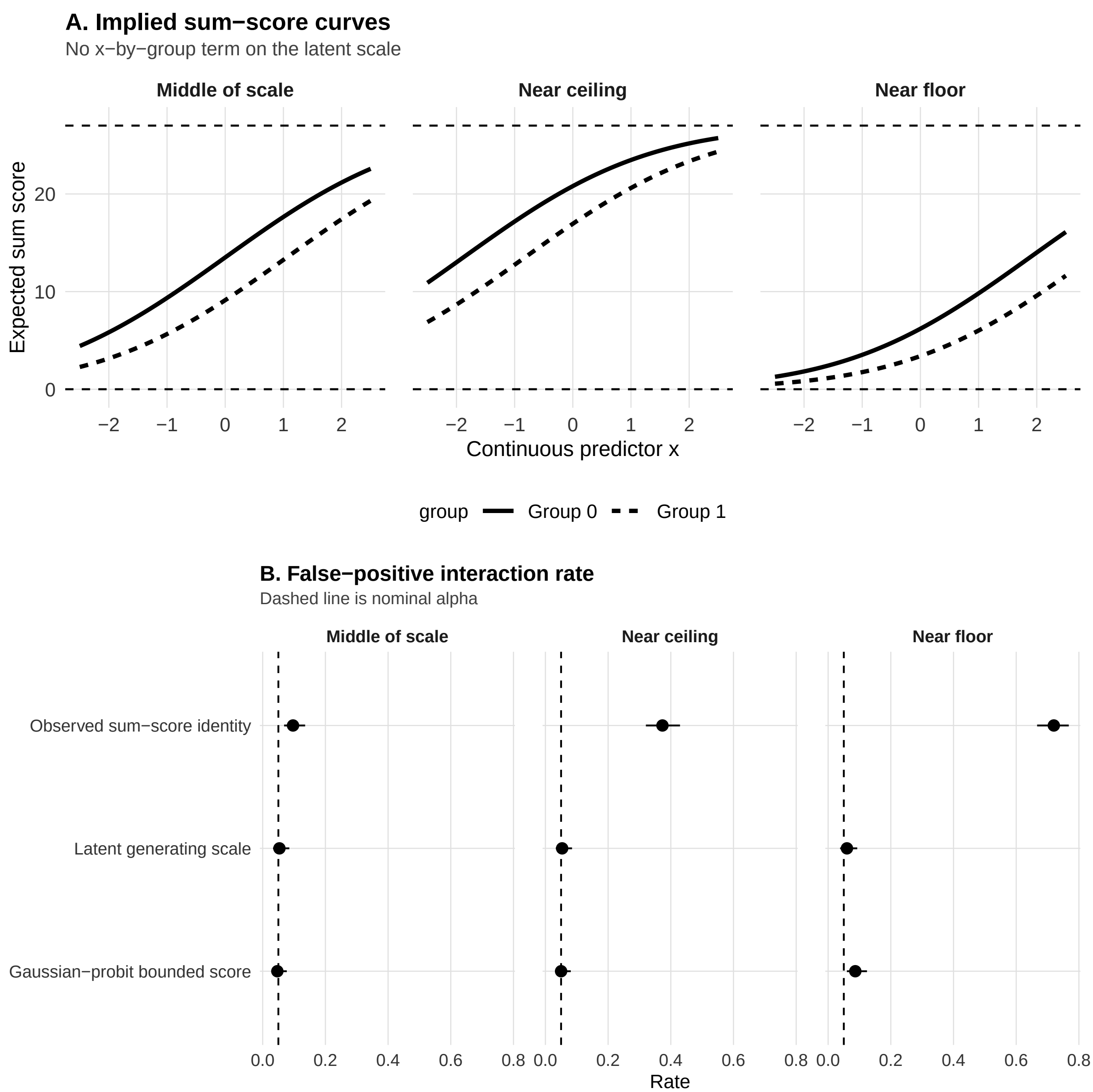

Figure 7

Simulation 3: within-family link choice in repeated-trial random-intercept GLMMs. Panel A shows the deterministic cell probabilities generated by additive logit or probit models at random intercept 0. Panel B shows the false-positive rate for the condition-by-group product term when the generating and fitted links are crossed. Error bars are Wilson 95% confidence intervals, and the dashed line marks nominal alpha.